

Araz Mehrabani

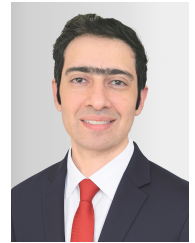
Mechanical Engineer | Design · FEA · Structural Dynamics

Herzogstraße 13-19, Frankfurt am Main

+49 1575 436 3238

araz.h.mehrabani@gmail.com

Date of birth: 28.07.1989



PROFILE & CORE COMPETENCIES

Mechanical engineer with several years of industrial experience across **mechanical design, product development, FE modeling and structural dynamics**. Hands-on work from component/assembly design, sizing and manufacturing drawings through FE model setup, static/modal/harmonic analysis, design revisions and experimental validation.

TECHNICAL FOCUS

Mechanical Design & CAD: CATIA V5, SOLIDWORKS; machine and drive concepts, components/assemblies, design revisions, 2D manufacturing drawings, dimensional tolerances, selective GD&T, shaft/bearing fits and BOMs.

FEA & Structural Dynamics: ANSYS Workbench; geometry reconstruction/simplification, line-body/beam-element modeling, materials, contacts, BCs/load cases, meshing, static/modal/harmonic analysis, resonance and operating-range assessment.

Engineering Calculations & Validation: material selection, shaft/spring calculations, mechanisms, gear ratio, torque/speed, motor/actuator selection, laboratory spring testing and result correlation.

Data & Prototyping: Python/MATLAB, NumPy/pandas/Matplotlib/SciPy, FFT/order analysis, time series; G-code, Arduino/Marlin and stepper calibration.

PROFESSIONAL EXPERIENCE

Development Engineer — Sabaniroo, Tehran

01/2022–02/2023

- Mechanically developed a two-axis PV-panel cleaning system in **SOLIDWORKS**: rail/wheel guidance, belt-driven carriage motion, torque/speed/ratio and wheel/rail-load calculations, motor selection, CAD and prototype testing; approx. **10% lower material use**.
- Performed an **ANSYS** assessment of a wind-turbine tower-to-foundation connection: model setup, material/contact definition, boundary conditions, meshing, and stress/deformation evaluation of critical regions.

Mechanical Engineer — TehranRigzar, Tehran

03/2017–12/2021

- Designed conveyors, jaw crushers, sand washers and vibrating screens in **CATIA V5**: parts/assemblies, drawings, BOMs, tolerances/selective GD&T, shaft/bearing fits, material selection, shaft/spring calculations and design revisions; supported welded-structure fabrication.
- Reconstructed the vibrating screen in **ANSYS** as a line-body/beam-element FE model and performed static, modal and harmonic analyses of stress, deformation, natural frequencies and mode shapes to define a resonance-safe operating range; calculated and experimentally validated the spring system.

Working Student – Mechanical Design — Ario Fidar Maham Co., Tehran

11/2012–05/2013

- Recreated a 40x90 jaw crusher in **CATIA V5**, including individual components, assemblies and manufacturing drawings.

SELECTED ENGINEERING WORK

Master's Thesis (ongoing) – Dynamic Load Analysis: Python-based time-series pipeline with filter comparison, FFT/order diagnostics and 0P/1P/2P harmonic analysis for dynamic structural-load response assessment.

Low-Cost Hand Prosthesis – Prototyping: CAD and 3D printing of a modular hand prosthesis; rebuilt a printer using Arduino/Marlin, hardware wiring, firmware modification and Slic3r process settings for ABS/PLA.

EDUCATION

M.Sc. Wind Energy Engineering, Hochschule Flensburg — current grade: 1.9

03/2023–11/2026

M.Sc. Mechanical Engineering, Hakim Sabzevari University — Control & Robotics

09/2014–01/2017

B.Sc. Mechanical Engineering, Islamic Azad University — Structural Dynamics

09/2008–06/2013

LANGUAGES

German: C1 (actively improving). English: C1 / IELTS 7.0. Persian: Native.